

Comparative Analysis of Churn Prediction Models

A Technical Evaluation Report

Prepared by: Data Science Team

Subject: Telco Customer Churn Strategy

Executive Summary

This report provides a detailed comparative analysis of three machine learning architectures for predicting customer churn. The primary business objective is to maximize the identification of at-risk customers (Recall for Churners) to enable proactive retention campaigns. We evaluate a baseline Random Forest classifier, a class-weighted Logistic Regression model, and an advanced Deep Learning architecture (Multi-Layer Perceptron). The analysis includes feature importance, confusion matrices, and ROC-AUC comparisons to offer a comprehensive view of model performance.

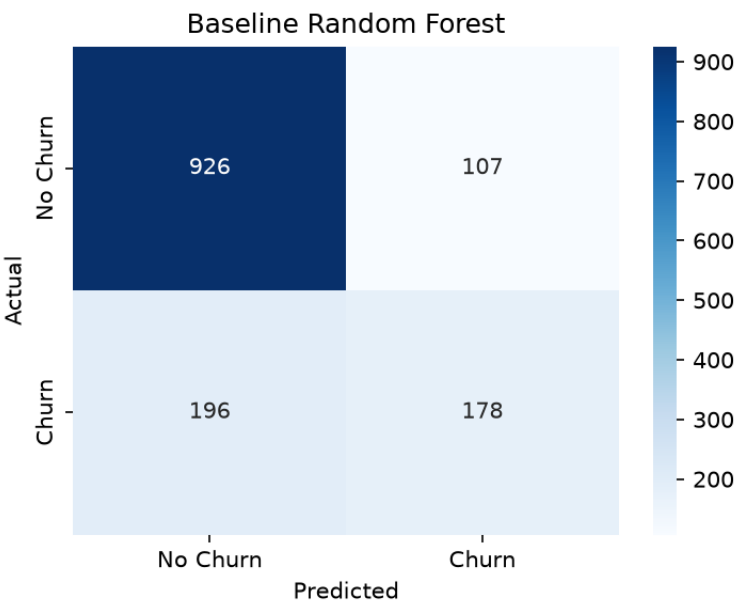
Model 1: Baseline Random Forest

Overall Accuracy: 78.46%

The baseline model achieves high accuracy but suffers from severe class imbalance, predicting 'No Churn' too aggressively.

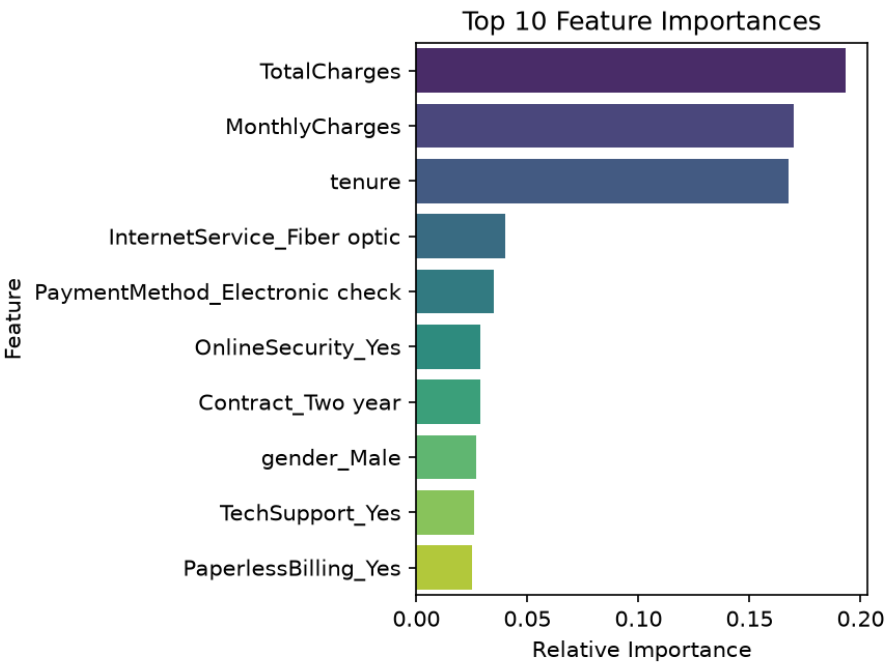
Churn Recall: Only 47.59% of actual churners were identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.83	0.90	0.86	1033.0
Class 1 (Churn)	0.62	0.48	0.54	374.0
Macro Avg	0.72	0.69	0.70	1407.0
Weighted Avg	0.77	0.78	0.77	1407.0



Feature Importance Analysis

The Random Forest model allows us to extract the most critical features driving customer churn. The chart below illustrates the top 10 factors influencing the model's decisions.



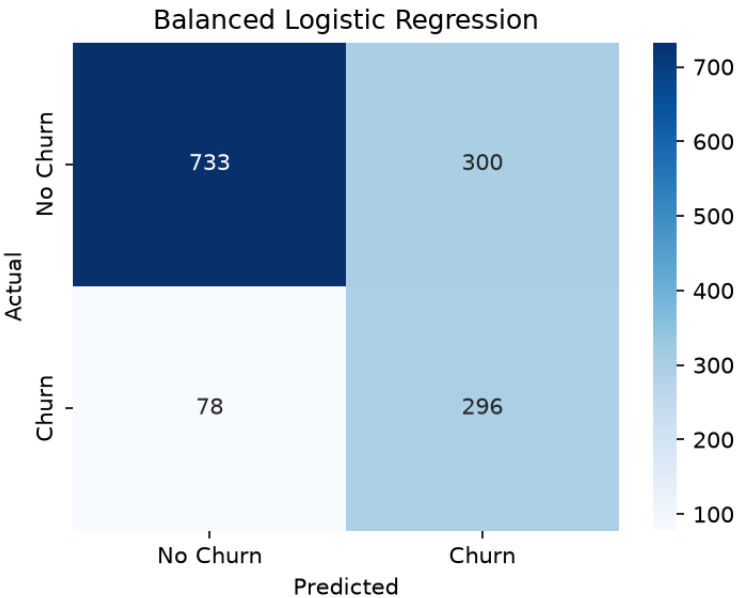
Model 2: Balanced Logistic Regression

Overall Accuracy: 73.13%

By applying 'balanced' class weights, this linear model heavily penalizes the misclassification of the minority class (Churners).

Churn Recall: 79.14% of actual churners were successfully identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.90	0.71	0.80	1033.0
Class 1 (Churn)	0.50	0.79	0.61	374.0
Macro Avg	0.70	0.75	0.70	1407.0
Weighted Avg	0.80	0.73	0.75	1407.0



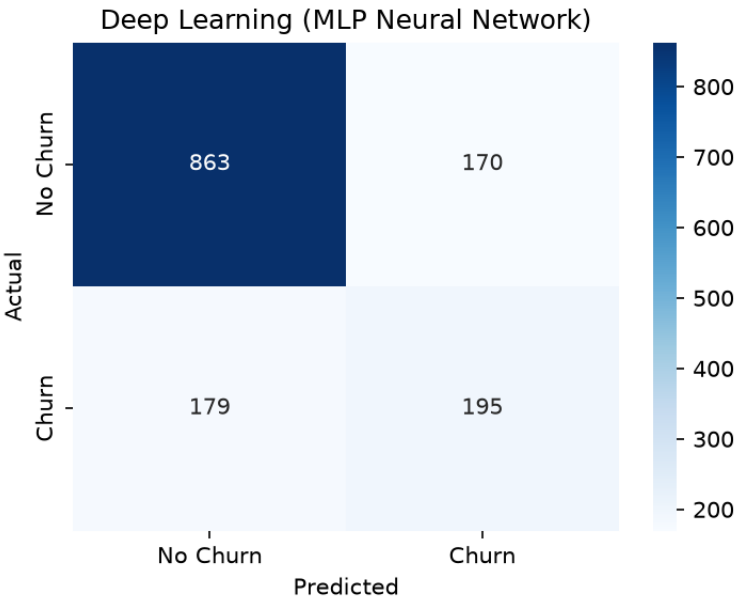
Model 3: Advanced Deep Learning (MLP)

Overall Accuracy: 75.20%

This advanced deep learning architecture utilizes multiple hidden layers to capture highly complex, non-linear feature interactions in the customer data.

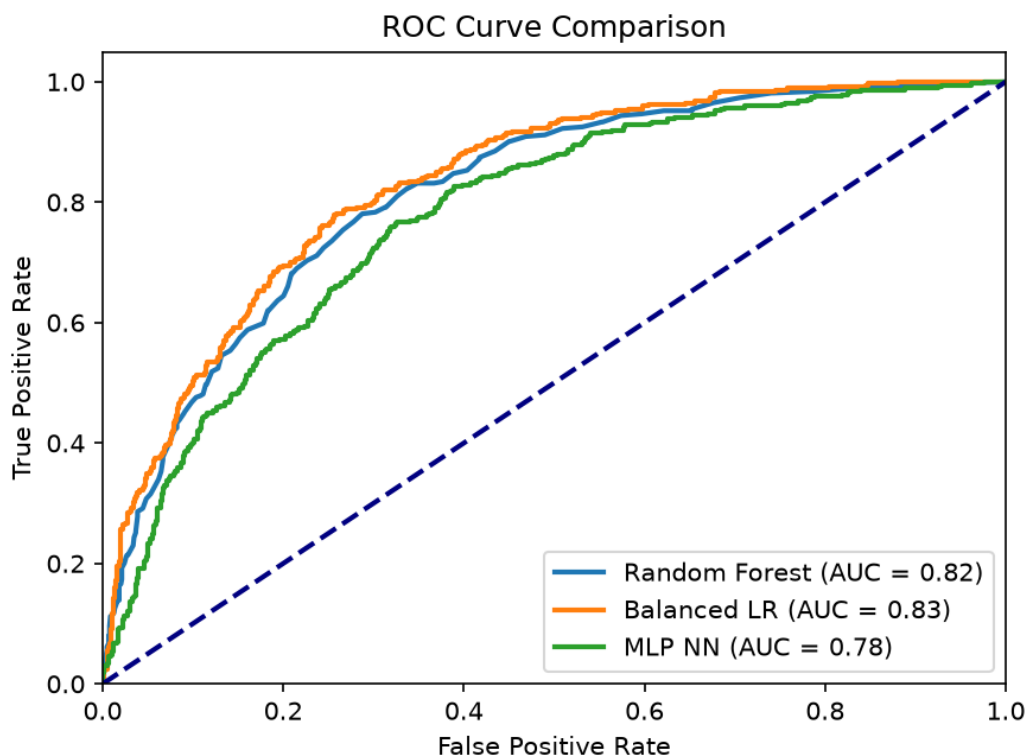
Churn Recall: 52.14% of actual churners were successfully identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.83	0.84	0.83	1033.0
Class 1 (Churn)	0.53	0.52	0.53	374.0
Macro Avg	0.68	0.68	0.68	1407.0
Weighted Avg	0.75	0.75	0.75	1407.0



Model Comparison: ROC-AUC

The Receiver Operating Characteristic (ROC) curve below compares the true positive rate versus the false positive rate across all three models at various threshold settings. The Area Under the Curve (AUC) summarizes the overall performance, where a value closer to 1.0 indicates a superior model.



Strategic Recommendation

The baseline Random Forest misses over half of the churning customers. While the Logistic Regression model captures 79.14% of churners, the Deep Learning model captures 52.14% of churners while maintaining a robust F1-score of 0.53. Depending on the specific cost of false positives vs. false negatives, either Logistic Regression or the Deep Learning model provides a massive upgrade over the baseline.